

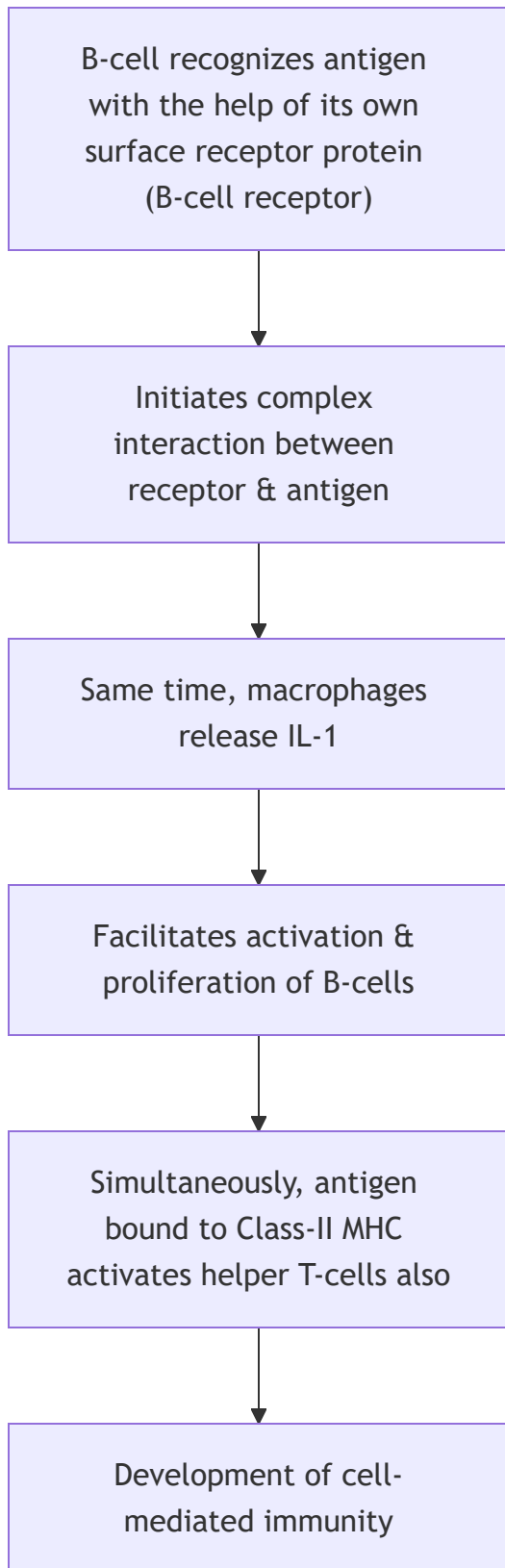
HUMORAL IMMUNITY

- The immunity mediated by antibodies (*by B-lymphocytes*).
- Also called B-cell immunity:
 - ↓
 - B-lymphocytes provide immunity through humors (*body fluids*).
- Antibodies (γ -globulins) produced by B-lymphocytes.
- Major defense mechanism against bacterial infection.

ROLE OF ANTIGEN PRESENTING CELLS

- Antigen-presenting cells present antigenic products:
 - ↓
 - Bound with HLA to B-cells
 - ↓
 - B-cells activated.

PRESENTATION OF ANTIGEN



TRANSFORMATION OF B-CELLS

- **Transformed into 2 types :—**
- a. Plasma cells
- b. Memory cells

ROLE OF PLASMA CELLS

- **Plasma cells destroy foreign organisms** by producing antibodies (*immunoglobulins*):
- ↓
- 2000 molecules of antibodies / second.
- **Antibodies released into lymph** → into circulation.
- **Antibodies produced until end of lifespan** of each plasma cell (*days – weeks*).

ROLE OF MEMORY B-CELLS

- These occupy lymphoid tissues throughout the body.
- **Memory cells are in inactive condition:**
- ↓
- Until body is exposed to same organism for 2nd time.
- ↓
- **Antibodies produced (2nd exposure) are more potent than (1st exposure).**
- ↓
- Phenomenon forms **basic principle of vaccination**.

ROLE OF HELPER T-CELLS

- **Activated helper T-cells secrete 2 substances:**
- a. IL-2
- b. B-cell growth factor

These substances promote :—

- a. Activation of more number of B-lymphocytes

- b. Proliferation of plasma cells
- c. Production of antibodies